

Please write clearly in block capitals.

Centre number

Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE**Paper 2 (Generated practice paper - Set 02)**

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours**Materials**

For this paper you may use:

- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
 - use good English
 - organise information clearly
 - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
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6	
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9	
10	
11	
TOTAL	

Answer all questions in the spaces provided.

0 1 . 1

Explain how the relationship between coral polyps and zooxanthellae allows a high biomass of life in nutrient-poor tropical seas.

[4 marks]

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0 1 . 2

Explain why growing coral reefs help to resist climate change.

[2 marks]

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0 1 . 3

Discuss the importance of tropical coral reefs and evaluate the methods used to conserve them.

[9 marks]

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Extra space

15

0 2 . 1

Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

0 2 . 2

Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

0 2 . 3

Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

5

0 3

Table 1 shows the catch made by a shrimp trawler during one fishing trip.

Table 1

Component of catch	Mass / tonnes
Shrimps (target species)	0.6
Crabs, sea urchins, molluscs and starfish	5.9
Immature fish of commercial species	1.9

0 3

. 1

Calculate the percentage of the total catch that was by-catch.

[2 marks]

Answer

%

0 3

. 2

Explain why shrimp trawling produces such a high proportion of by-catch.

[1 mark]

0 3

. 3

Explain why k-selected species such as the common skate and the Greenland shark are especially vulnerable to overfishing.

[3 marks]

0 3

. 4

Evaluate the methods that can be used to reduce the environmental impacts of fishing.

[9 marks]

[illegible]

Extra space

15

0 4 . 1 State what is meant by **hybrid vigour**.

[1 mark]

0 4 . 2 Explain how cross-breeding Zebu and Ayrshire cattle produced a breed suited to hot climates.

[2 marks]

0 4 . 3 Give **two** potential applications of cloning to livestock farming.

[2 marks]

1

2

5

0 5

Table 1 compares nitrogen fixation by Rhizobium bacteria in legume root nodules with the Haber process.

Table 1

Feature	Biological fixation	Haber process
Energy source	Sunlight captured by the plant (carbohydrates)	Fossil fuels (natural gas / coal)
Temperature	Soil temperature (10-30 °C)	About 450 °C
Pressure	Atmospheric	200 atmospheres
Catalyst	Nitrogenase enzyme	Iron
Energy density of input	Low	High
Waste products	None - all products used	CO ₂ , NO _x

0 5

. 1

Use Table 1 to explain how natural processes differ from human industrial processes in their use of energy.

[4 marks]

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0 5

. 2

Give **three** other natural processes driven by solar energy that human society depends on.

[3 marks]

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0 5 . 3

Explain how the principle of low-temperature natural processes is being applied in human systems.

[3 marks]

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10

0 6 . 1

Which feature of the Earth prevents most atmospheric gases from escaping into space? Tick **one** box.

[1 mark]

- | | |
|-----------------------------|--------------------------|
| A Its mass | ☐ |
| B Its magnetic field | ☐ |
| C Its distance from the Sun | ☐ |
| D Its speed of rotation | ☐ |

0 6 . 2

Explain how the speed of rotation of the Earth and the tilt of its axis affect the conditions for life.

[2 marks]

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06 . 3

Explain why the presence of liquid water on Earth depends on **both** the distance from the Sun **and** atmospheric pressure.

[2 marks]

5

07 . 1

Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

A Radio collar

☐

B Acoustic / satellite tag

☐

C Pitfall trap

☐

D Geolocator tag recovered by recapture

☐

07 . 2

Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

5

08

The Rio Bravo Conservation and Management Area in Belize exploits mahogany under a sustainable management plan while conserving a rainforest of over 200 tree, 400 bird and 70 mammal species.

08

1

Describe **five** features of the timber harvesting at Rio Bravo that make it sustainable.

[5 marks]

1

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2

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08

2

Explain why involving local communities is essential to the success of the reserve.

[3 marks]

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08

3

Explain why selective logging is better for wildlife than clear-felling, and give **one** disadvantage.

[2 marks]

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10

09

A new pond was dug in a nature reserve. Table 1 shows the vegetation recorded at intervals over 60 years.

Table 1

Years after creation	Dominant vegetation	Water depth at centre / m
1	Single-celled algae, floating duckweed	2.0
5	Submerged pondweeds; reeds at margins	1.8
15	Reeds and reedmace over most of pond; water lilies	1.0
30	Marsh plants, sedges; willow and alder at edges	0.2
60	Willow / alder carr; oak seedlings	0 (soil waterlogged)

09

1

Name the type of sere shown in Table 1 and identify the pioneer species.

[2 marks]

09

2

Explain the changes shown in Table 1.

[5 marks]

09 . 3

Explain why a lithosere and a hydrosere in the same region end with the same climax community.

[2 marks]

09 . 4

Describe the early stages of a lithosere.

[1 mark]

10

10 . 1

Explain why the potential to increase food supplies by cultivating new land is limited.

[2 marks]

10 . 2

Explain why importing food may reduce the availability of food in the area where it is produced.

[2 marks]

10 . 3

Explain why organic agriculture is more than simply 'not using pesticides and artificial fertilisers'.

[2 marks]

10 . 4

Evaluate the strategies that could be used to make agriculture more sustainable.

[9 marks]

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Write an essay on one of the following topics.

1	1	.	1
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Fishing is the last large-scale human hunting activity. Discuss the environmental impacts of fishing and evaluate the methods used to make the exploitation of fish populations sustainable.

[25 marks]

OR

1 1 . 2

Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

Do not write outside the box

A large rectangular box containing 30 horizontal dotted lines for writing.

[illegible]

END OF QUESTIONS